

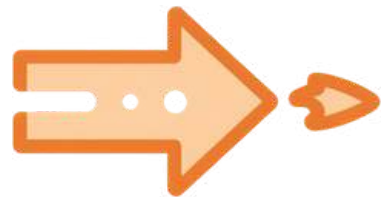


THE BALLOON ROCKET ADVENTURE: TO EVERY ACTION, AN OPPOSITE REACTION



THE SRIHARIKOTA COUNTDOWN

- Vihaan and Zarawar watch the Chandrayaan launch with Dada-ji at Sriharikota.
- Think about it:
- "Fire points down, rocket flies up. Why?"



Key Question:
How does pushing down
make rockets go up?

WHAT IS NEWTON'S 3RD LAW?

Definition:

For every push one way, there's a push back the opposite way!

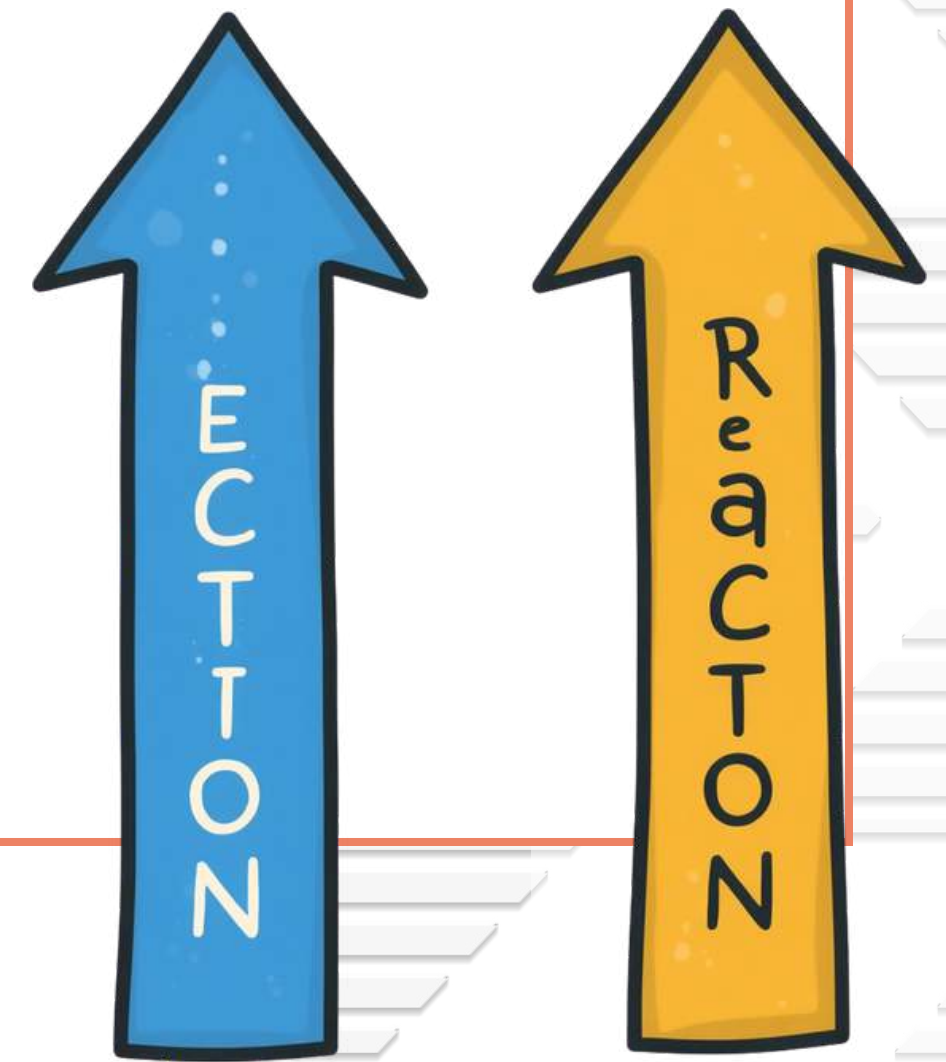
Key concept: Action and Reaction

When you push something, it pushes you back with the same strength!

Remember:

Action = The first push

Reaction = The push back



MEET THE BALLOON ROCKET

Our rocket is a balloon flying on a string!

Balloon = Rocket

Straw = Track

String = Path



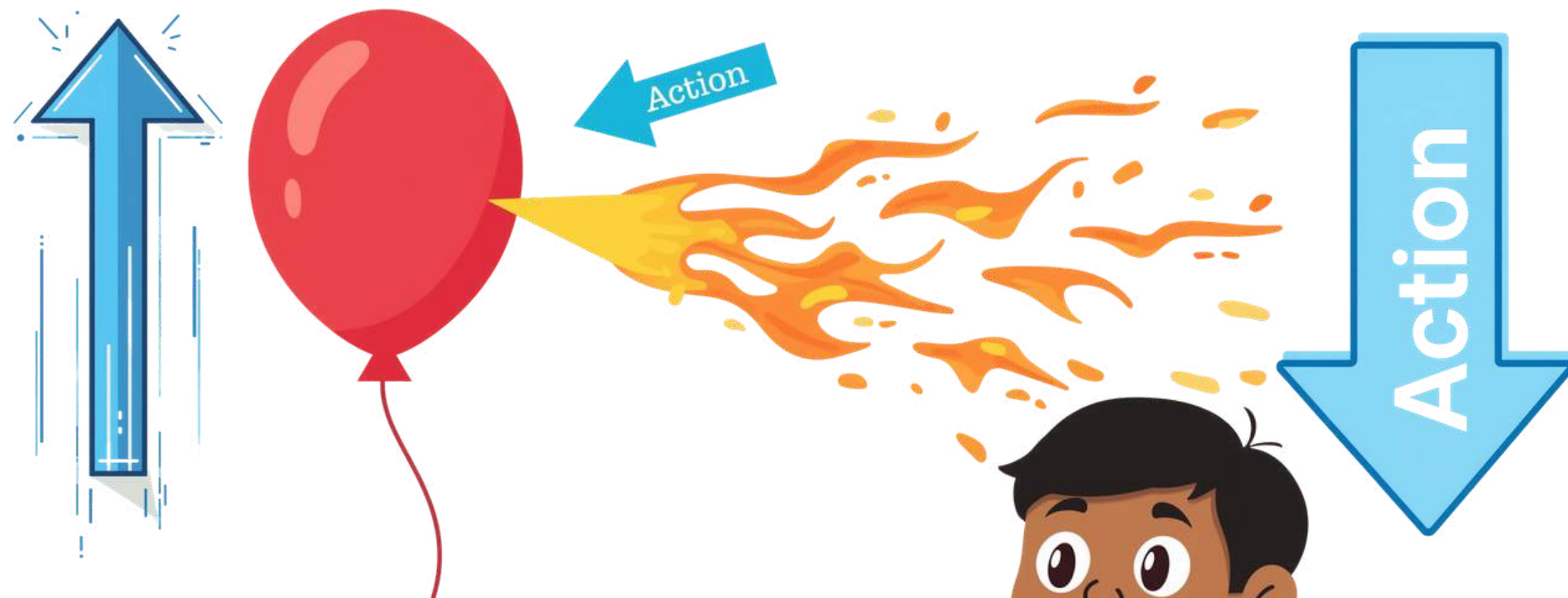
What You Need:
One balloon
One straw
One long string

ACTION PUSH: AIR RUSHES BACK

When we let go, air rushes backward out of the balloon.

Key idea:

This backward push of air is called the ACTION force!



Action Force:
Air rushes OUT
Pushes BACKWARD

REACTION PUSH: BALLOON FLIES FORWARD

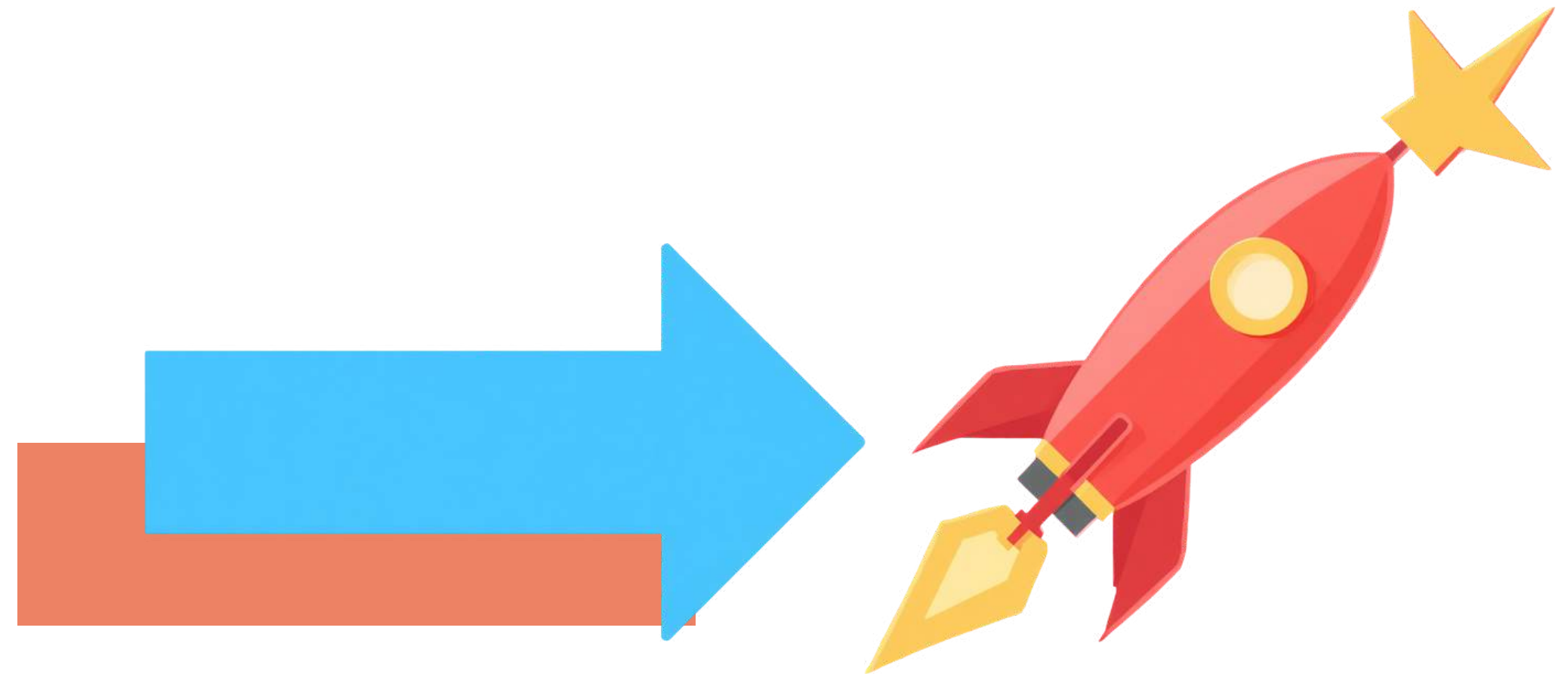
- The balloon zips forward because the air pushes back!
- Key idea:
- This is the Reaction force — the opposite push!



Reaction Force:
Balloon moves RIGHT
Air pushed LEFT

ACTION AND REACTION TOGETHER

- Air pushes back
- (Action)
- Blue arrow pointing left
- Action and Reaction always happen in pairs, opposite directions.
- Balloon zips forward
- (Reaction)
- Gold arrow pointing right
- They work together!



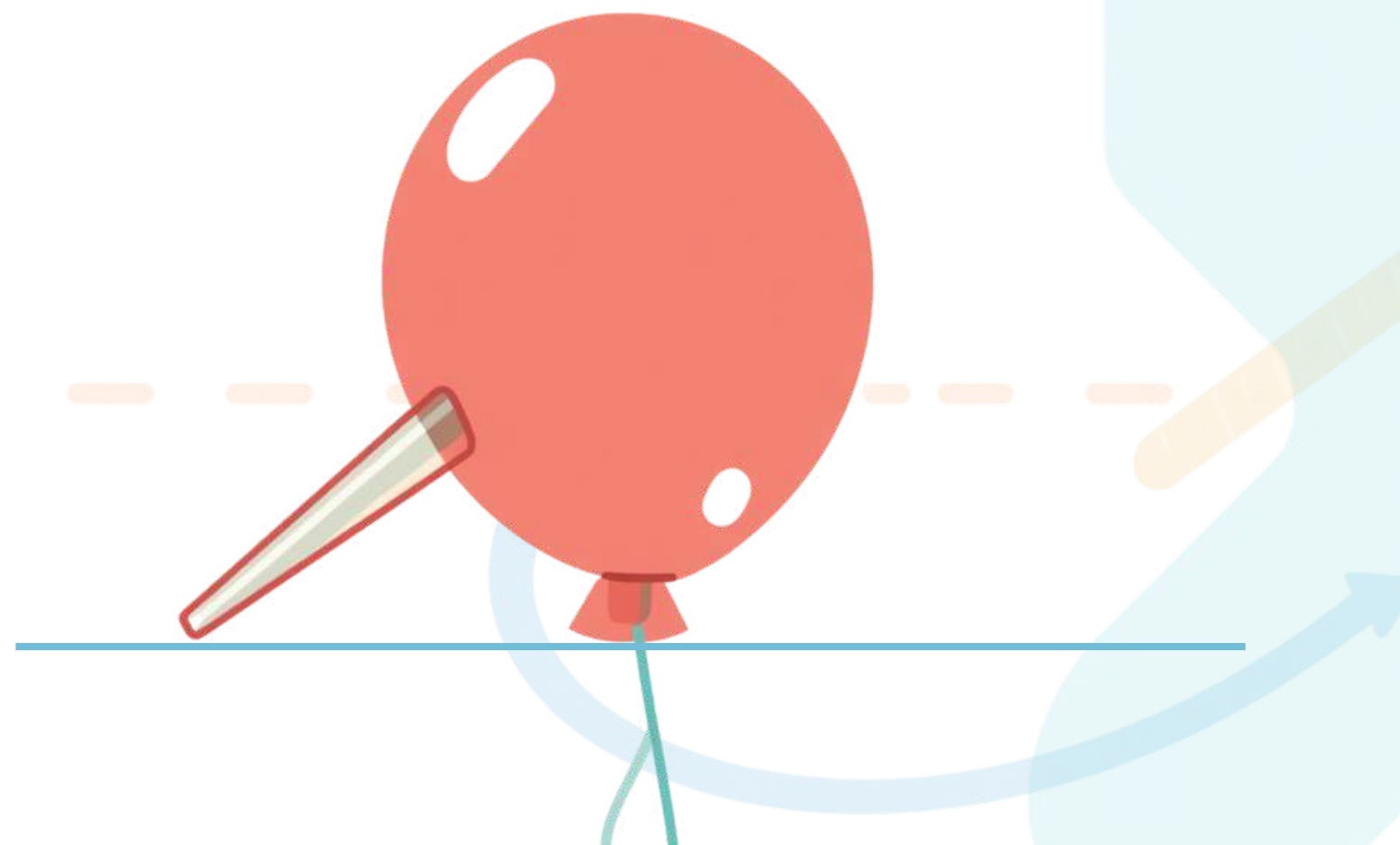
ISRO LAUNCH DIRECTOR SIMULATOR - TRIAL 1

Mission: You are the Launch Director! Your job is to fix the rocket's path.

Problem:

Nozzle points sideways → rocket swerves off path!

The air pushes sideways, so the balloon flies crooked.



Quick Check:
Which way should the nozzle point for a straight flight?

ISRO LAUNCH DIRECTOR SIMULATOR - TRIAL 2

Problem: Less air pushes weakly → rocket moves slowly

What happens:

A half-inflated balloon has less air to push out

Weak action force = weak reaction force = slow rocket!



Quick Check:
Why does more air make the rocket faster?

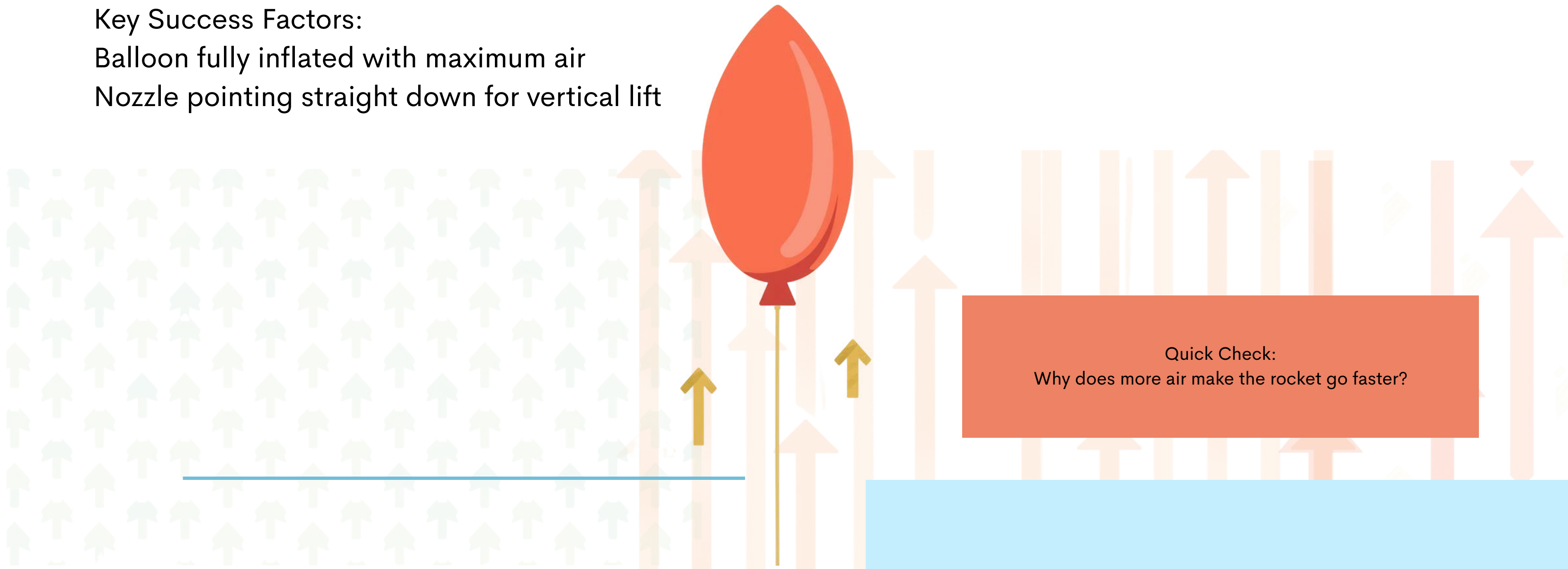
ISRO LAUNCH DIRECTOR SIMULATOR - TRIAL 3

Perfect Launch: Max air + straight nozzle → perfect vertical rocket flight!

Key Success Factors:

Balloon fully inflated with maximum air

Nozzle pointing straight down for vertical lift

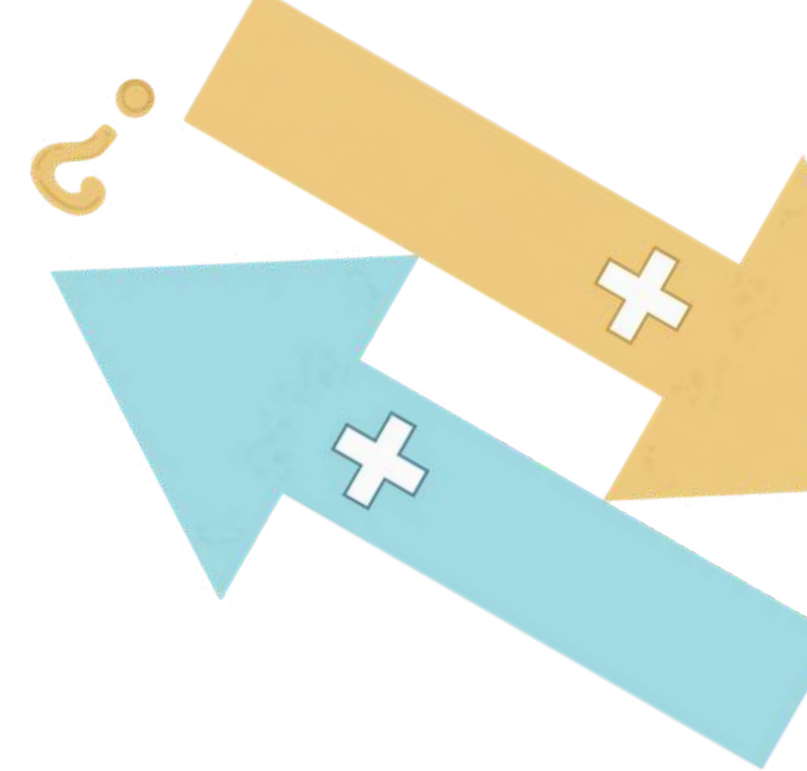
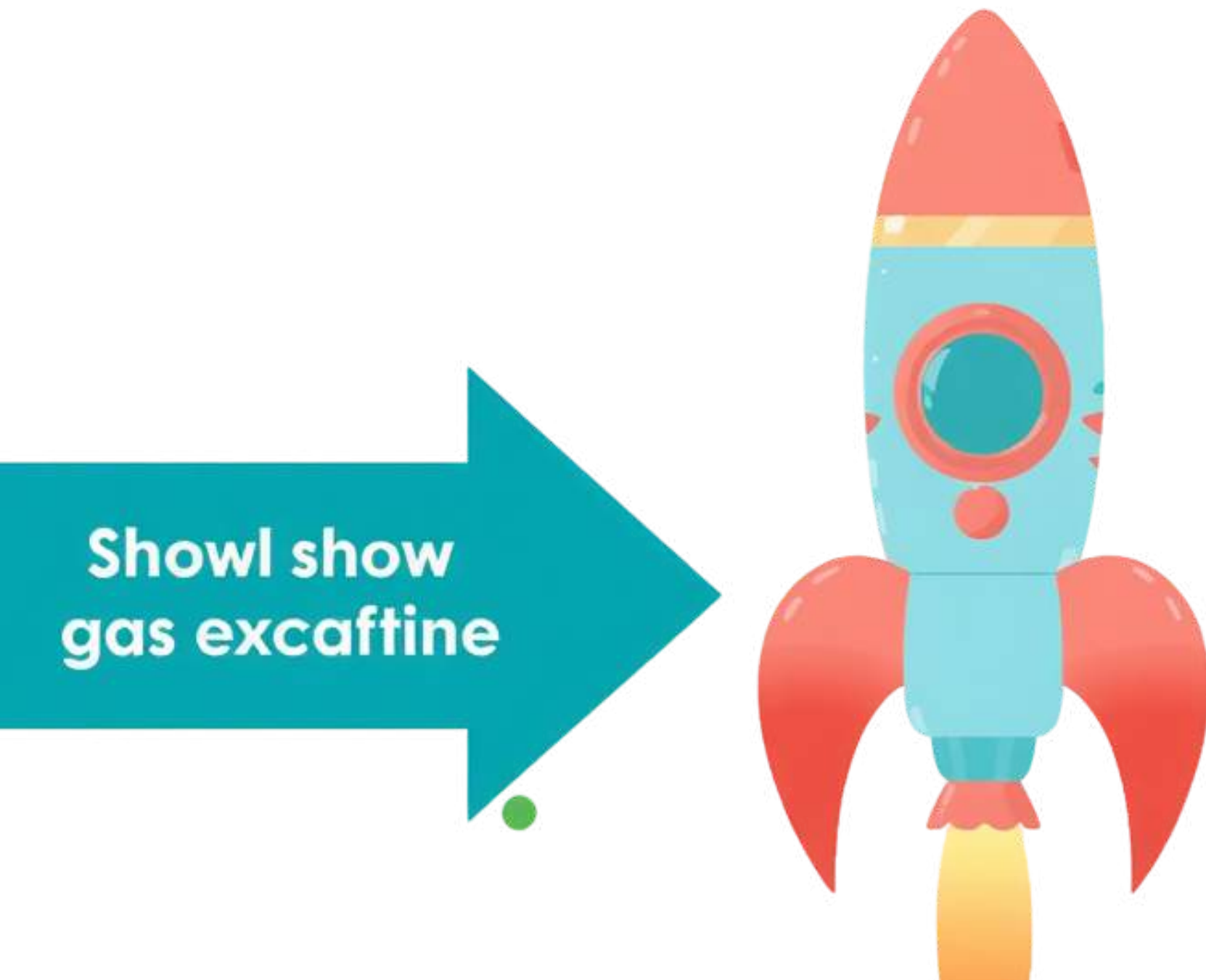


Quick Check:
Why does more air make the rocket go faster?

QUICK QUIZ: WHICH WAY DOES THE ROCKET FLY?

If rocket gas pushes LEFT, which way does the rocket fly?

- A) Left (same direction as the gas)
- B) Right (opposite direction to the gas)

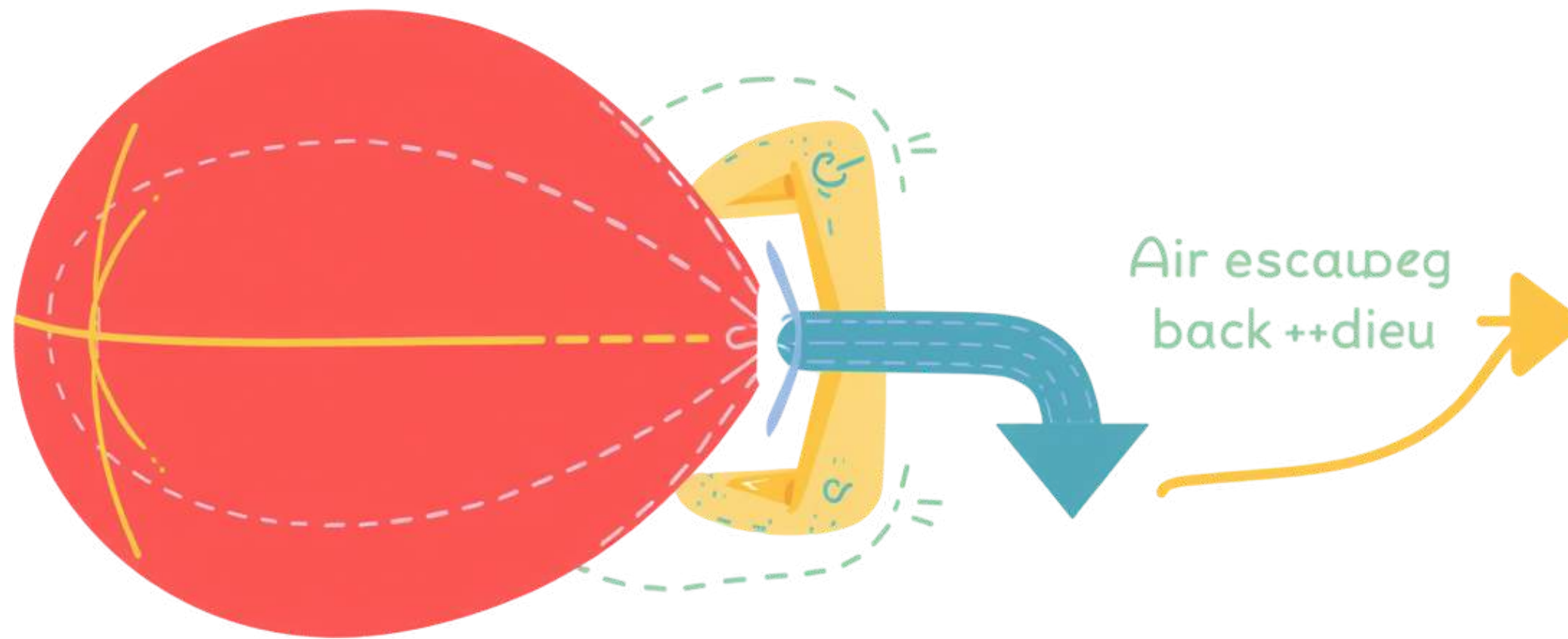


Think About It:
Remember the Double Push Rule!

QUICK QUIZ: IDENTIFY THE ACTION FORCE

What is the "Action" force in our balloon rocket?

- Air rushing backward
- Balloon flying forward
- String pulling
- Straw moving



Think About It:
Which one starts the motion?

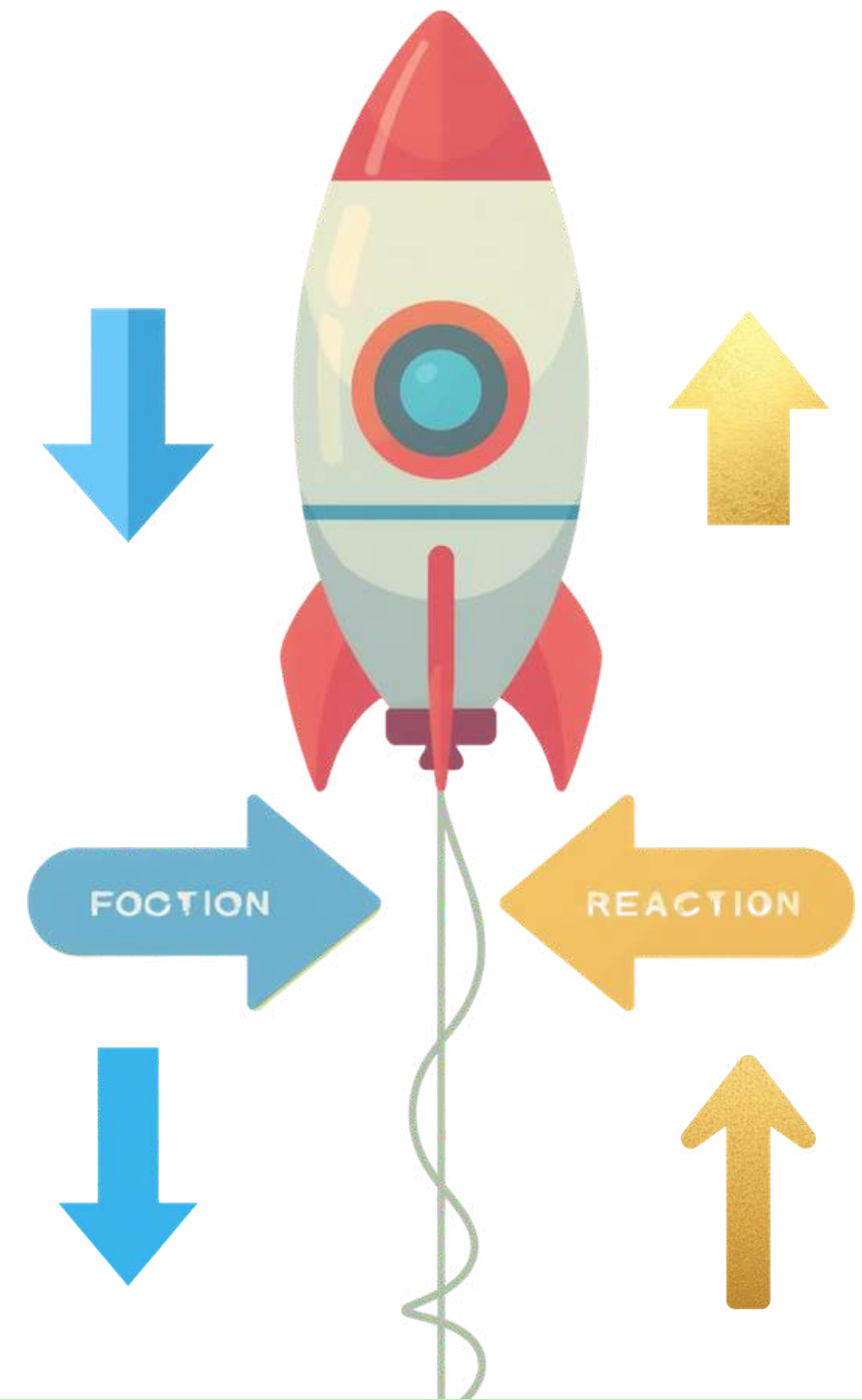
QUICK QUIZ: MATCH THE TERMS

Instructions:

Match each term to the correct arrow!

Action → Blue Arrow (Air pushes backward)

Reaction → Gold Arrow (Balloon zips forward)



Your Turn:
Drag Action and Reaction to
the right arrows!

THE DOUBLE PUSH RULE

Key Takeaways:

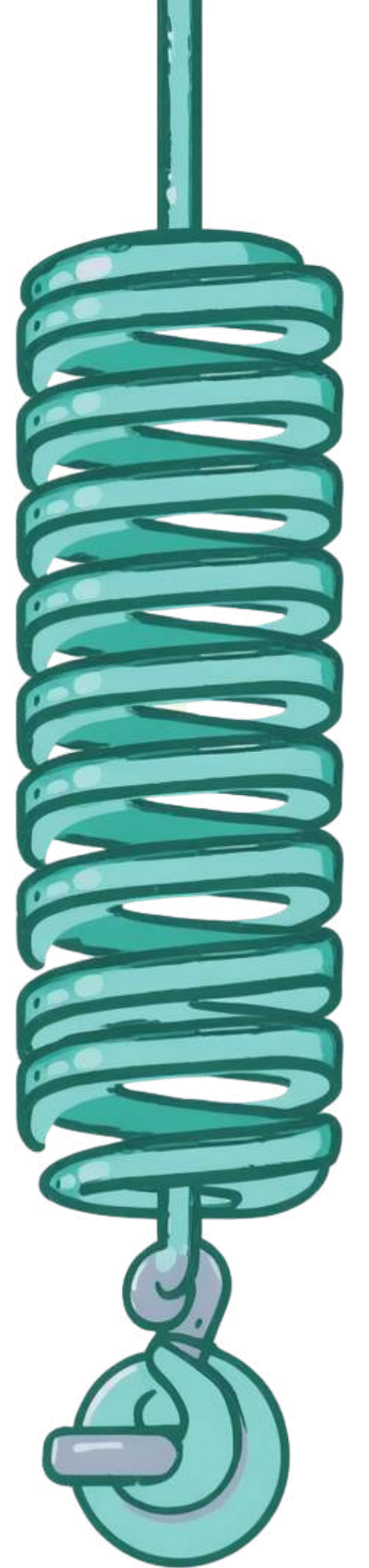
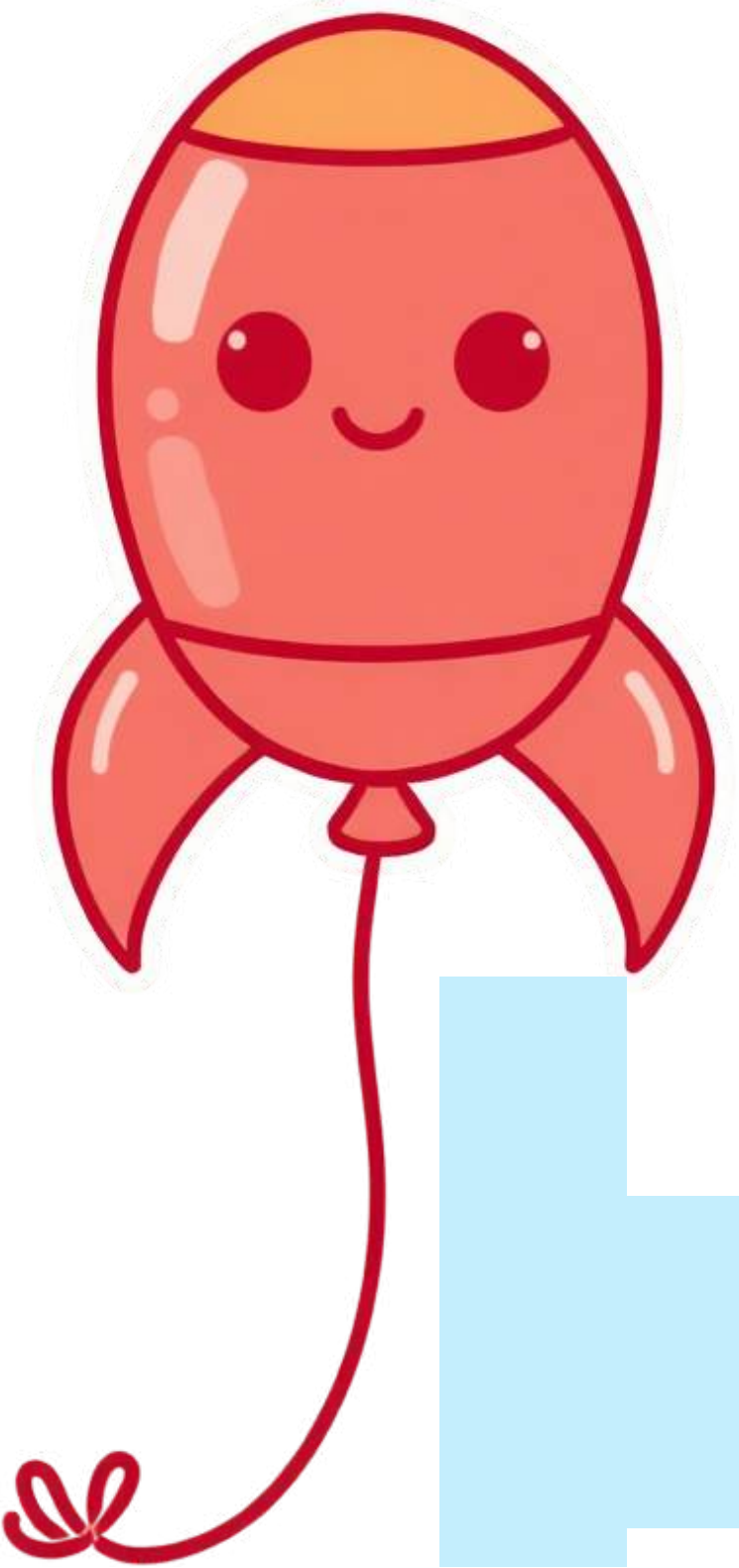
Forces always come in pairs, pushing in opposite directions

Balloon rocket shows action (air rushes back) and reaction (balloon zips forward)

Changing the nozzle direction changes the rocket's path

ISRO's LVM3 rocket pushes gas down to fly up into space

Feel proud of India's amazing space missions!



HOME QUEST: BE A PHYSICS EXPLORER!

Instructions: Stand on a skateboard or roller skates.
Push a wall forward. Watch your body roll back!

Your Mission:

Draw the two opposite pushes you feel.
You are a real Launch Director now!



Try This:
Push wall forward
You roll backward



THANK YOU, PHYSICS EXPLORERS!
JAI HIND!



Keep dreaming big like
ISRO! See you next session!